

Software Design Document

To design a URL shortener system, we need to consider the following components:

- **URL Receiver**: receives the original URL from the user
- **URL Shortener**: generates a unique short URL
- **Storage**: stores the mapping between the original URL and the short URL
- **URL Redirector**: redirects the user to the original URL when the short URL is accessed
- **API**: handles user requests and returns responses

Here is a high-level architecture for the system:

[URL Receiver] as Receiver

[URL Shortener] as Shortener

[Storage] as Storage

[URL Redirector] as Redirector

[API] as API

Receiver --> API

Shortener --> Storage

API --> Shortener

API --> Redirector

Redirector --> Storage

Storage --> Redirector